

Introduction to SQL



Data Science Landscape



- **Structured Query Language (SQL)**
- The standard for **relational database management systems (RDBMS)**
 - A database management system that manages data as a collection of tables in which all relationships are represented by common values in related tables

Introduction to SQL

- SQL Functions:

1. Data Definition Language (DDL)

- Create database objects:
 - Tables,
 - Indexes,
 - Views
- Define access rights to those database objects

2. Data Manipulation Language (DML)

- Commands to:
 - Insert,
 - Update,
 - Delete,
 - Retrieve data

Introduction to SQL

- SQL Functions:

3. Data Control Language (DCL)

- Grant
- Revoke

4. Transaction Control Language (TCL)

- Commit
- Rollback
- Savepoint

- **Applications:**

- To **access data** in the RDBMS.
- To **describe** the data.
- To **define** the data in a database and **manipulate** that data.
- To **embed** within other languages using SQL modules, libraries & pre-compilers.
- To **create** and **drop** databases and tables.
- To create **view**, **stored procedure**, **functions** in a database.
- To set **permissions** on tables, procedures and views.

- **Applications:**

- Any program that needs to **store, manipulate** and **retrieve** data:
 - Webpages
 - Computer Applications
 - Mobile Applications
- Used by many systems:
 - MS Access, Oracle, Sybase, Informix, ...

SQL Overview

- Applications:





Welcome

email

password

☒ Remember Me

[SIGN IN](#)

[SIGN UP](#)

Sign Up

name

email

password

confirm password

[SIGN UP](#)

[SIGN IN](#)

← Meal options

[-] Additions

- ☐ Pickles
- ☒ Tomato
- ☐ Lettuce
- ☐ Cheese

← Settings

Phone ringtone

- ☐ None
- ☒ Callisto
- ☐ Ganymede
- ☐ Luna

[Install](#) & 1 MORE 



- ☒ [Install](#) 
- ☒ [Session_Start](#) 
- ☐ [clan_war_video](#)
- ☐ [SessionEnd](#)

SQL Environment

- **Catalog**

- A set of schemas that constitute the description of a database

- **Schema**

- The structure that contains descriptions of objects created by a user
- e.g. base tables, views, constraints.

- **DDL**

- Commands that define a database, including creating, altering, and dropping tables and establishing constraints

- **DML**

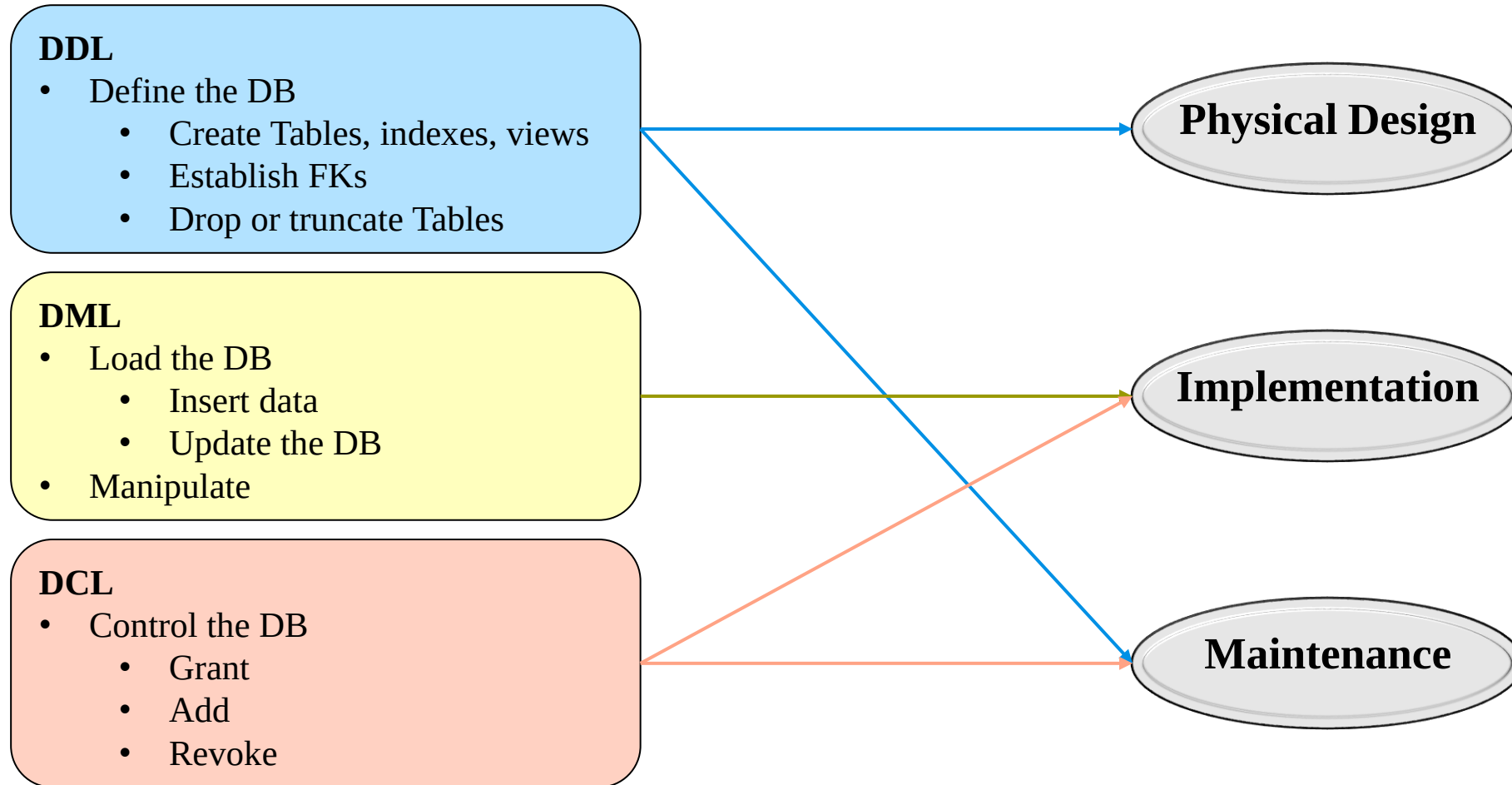
- Commands that maintain and query a database

- **Data Control Language (DCL)**

- Commands that control a database,
- e.g. administering privileges and committing data

SQL Environment

- DDL, DML, DCL, and the database development process



SQL Standard

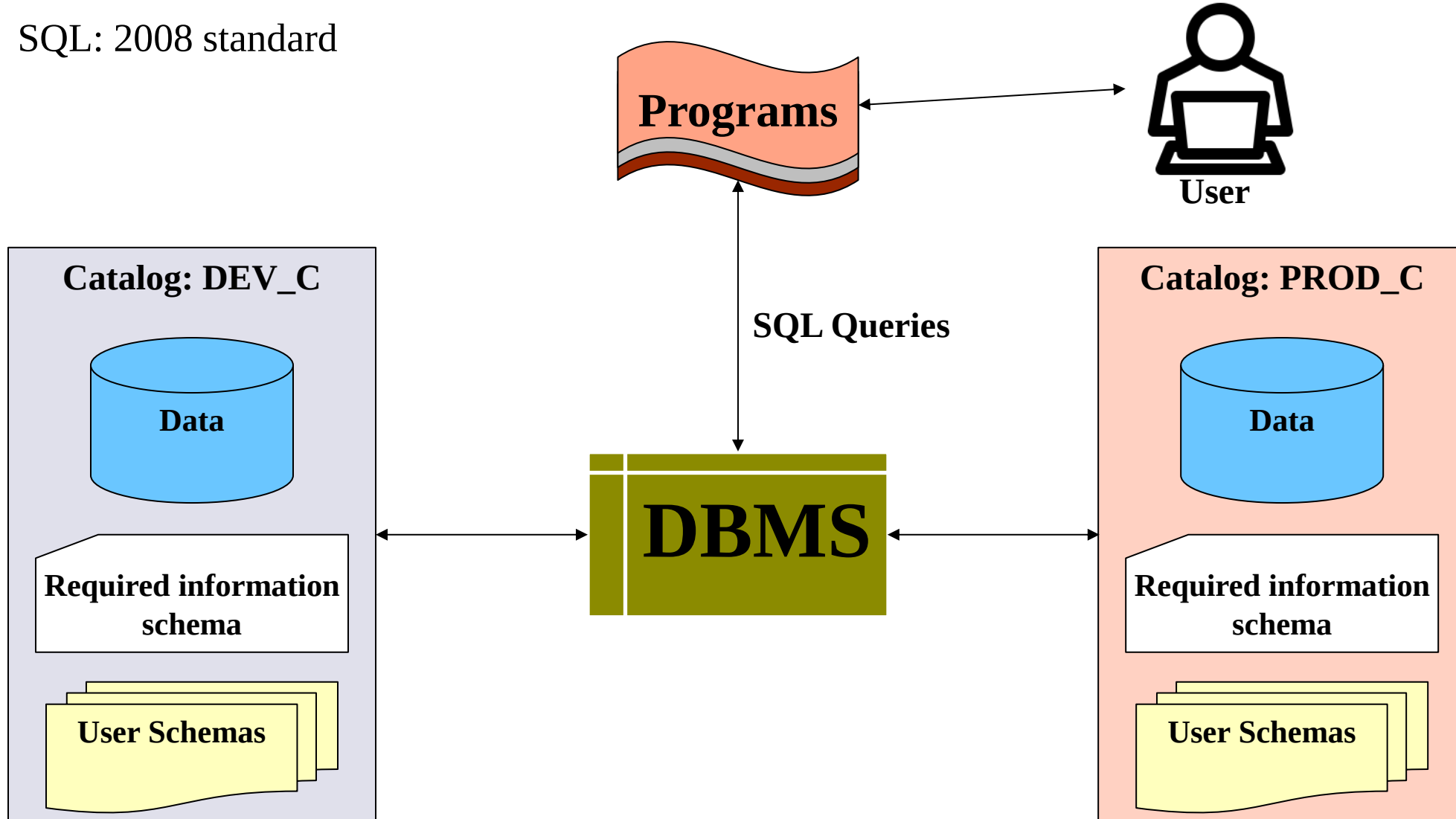
- Specify **syntax/semantics** for data definition and manipulation
- Define **data structures** and basic operations
- Enable **portability** of database definition and application modules
- Specify **minimal** (level 1) and **complete** (level 2) standards
- Allow for later growth/enhancement to standard (referential integrity, transaction management, user-defined functions, extended join operations, national character sets)

Standardized Relational Language

Advantages:

- Reduced training costs
- Productivity
- Application portability
- Application longevity
- Reduced dependence on a single vendor
- Cross-system communication

- **SQL environment,**
 - SQL: 2008 standard



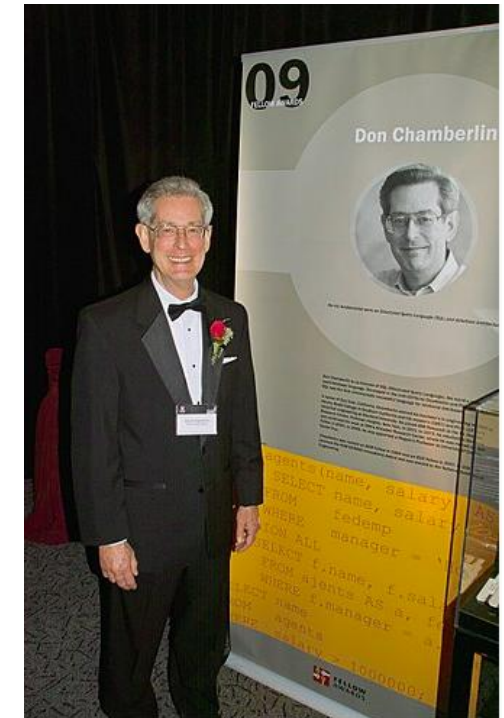
History of SQL

- **1970**: E. F. Codd develops **relational database** concept.
- **1974-1979** : System **R** with Sequel (later SQL) created at IBM Research Lab.
- **1979** : Oracle markets first **relational** DB with SQL.
- **1981**: SQL/DS first available **RDBMS** system on DOS/VSE.
- Others followed: **INGRES** (1981), **IDM** (1982), **DG/SGL** (1984), **Sybase** (1986).
- **1986**: **ANSI** SQL standard released.
- **1989, 1992, 1999, 2003, 2006, 2008**: Major **ANSI standard** updates.
- **Current**: SQL is supported by most major database vendors.

History

- IBM Sequel language developed as part of System R project at the IBM San Jose Research Laboratory
- Renamed as “Structured Query Language” (SQL)
- ANSI and ISO standard SQL:
 - SQL-86
 - SQL-89
 - SQL-92
 - SQL:1999 (language name became Y2K compliant!)
 - SQL:2003
- Commercial systems offer most, if not all, SQL-92 features, plus varying feature sets from later standards and special proprietary features.
 - Not all examples here may work on your particular system.

- **Donald D. Chamberlin**
 - Designed the original SQL
 - PhD from Stanford University
 - A researcher in the field of DB languages and systems
 - A member of System R Research Team



Types

- MySQL SQL
- MariaDB SQL
- PostgreSQL SQL (PL/pgSQL)
- SQLite SQL
- Microsoft SQL Server SQL (T-SQL)
- Oracle SQL (PL/SQL)
- IBM DB2 SQL
- Snowflake SQL
- Google BigQuery SQL
- Amazon Redshift SQL
- Sybase SQL (T-SQL)
- Firebird SQL



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